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Chapter 4: Bioenergetics

Bioenergetics: *The quantitative study of energy conversions and energy relationships in biological systems. Its most fundamental processes are photosynthesis and respiration (and photorespiration).*

4.1 Photosynthesis

- *Chemically a **redox** process: CO_2 (oxidized carbon) is reduced to glucose; water is the reducing agent and is oxidized to oxygen.*
- *Bio-energetically: energy poor molecules (CO_2, H_2O) are changed into energy rich glucose; the extra energy is absorbed from sunlight by photosynthetic pigments.*

Overall equation: $6CO_2 + 12H_2O$ --(light, chlorophyll)--> $C_6H_{12}O_6 + 6O_2 + 6H_2O$

SLO 1: Role of light, carbon dioxide and water

- ***Light:*** *sunlight is the ultimate energy source, an electromagnetic form of energy. **Electromagnetic spectrum** = full range of electromagnetic radiation. Visible light = 380 to 750 nm, effective for photosynthesis.*
- ***Action spectrum:*** *graph showing effectiveness of different wavelengths for photosynthesis (rate of photosynthesis in each colour). Blue (430 nm) and red (670 nm) are most effective.*
- ***Carbon dioxide:*** *the carbon source (raw material) used in the light independent reaction (Calvin cycle); plants are autotrophs. Air has 0.03-0.04% CO_2. Rate rises with CO_2 but above 1% it falls rapidly due to stomatal closing. Aquatic plants use dissolved CO_2, bicarbonates and carbonates.*
- ***Water:*** *raw material; hydrogen and electron donor. In light dependent phase it splits: $2H^+$, $2e^-$ and $1/2\ O_2$ (**photolysis of water**). Electrons refill P680 of PS II;*

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H^+ and e^- are taken by $NADP^+$ to make NADPH; oxygen is released.

- ***Van Niel (1931):** plants split water as a source of hydrogen, releasing oxygen (from bacteria that do not release O_2). **Confirmed 1940** with heavy isotope ^{18}O: water tagged with ^{18}O gave radioactive oxygen; CO_2 tagged with ^{18}O gave oxygen with no isotope, so O_2 comes from water.*

SLO 2, 3: Photosynthetic pigments and their roles

***Pigment:** Any substance that absorbs light energy for use in food synthesis. In plants pigments are embedded in thylakoid membranes (grana lamellae); in bacteria in cytoplasm.*

Chlorophylls	Carotenoids
Absorb mainly violet, blue, orange and red; green and yellow reflected	Terpenoid lipids; yellow, orange, red, brown; absorb strongly in blue-violet
Chl a: bluish green, in all photosynthetic organisms except bacteria	Accessory pigments with Chl b: absorb and transfer energy to Chl a
Chl b: yellowish green, absent in brown and red algae and bacteria	Protect Chl a from excess light; attract insects and birds for pollination, dispersal
Algae also have Chl c, d, e; bacteria have bacteriochlorophyll	Two types: carotenes (orange-red, e.g. beta carotene) and xanthophylls (yellow, e.g. lutein)

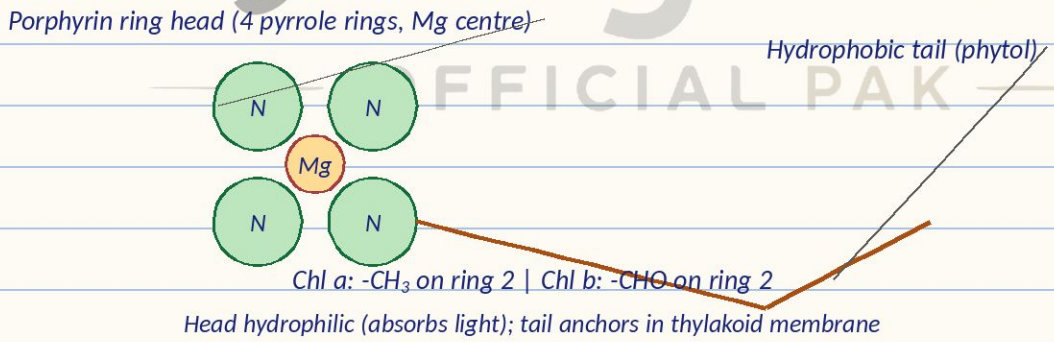


Fig: Structure of chlorophyll

- *Chl a = $C_{55}H_{72}O_5N_4Mg$; Chl b = $C_{55}H_{70}O_6N_4Mg$. Head: porphyrin ring of four pyrrole rings with Mg in centre (hydrophilic, light absorbing, on thylakoid surface). Two tails: phytol and methyl ester (hydrophobic, anchor in*

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*membrane). **Only difference:** Chl a has methyl (-CH_3) on ring 2, Chl b has aldehyde (-CHO).*

- *Chlorophyll synthesis needs sunlight, nitrogen, Mg^{2+} and warm temperature. Deficiency = **chlorosis**. Leaves turn yellow in autumn / winter because chlorophyll breaks down and carotenoids (lutein) show.*
- ***Roles:** primary role = absorb radiation and pass photon energy to the reaction centre; Chl a converts light energy to chemical energy (ATP and NADPH). **Accessory pigments** (Chl b, xanthophyll, carotenes) only absorb and transfer energy to Chl a, widening the spectrum used. Plants look green because green is reflected.*

SLO 4: Absorption spectra of chlorophyll a and b

***Absorption spectrum:** Graph showing relative absorption of different wavelengths of light by a particular pigment (measured with a spectrophotometer).*

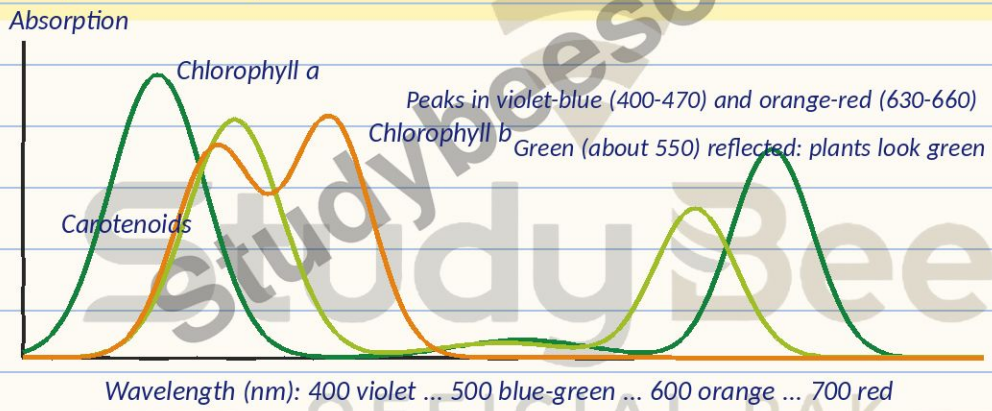


Fig: Absorption spectra of photosynthetic pigments

- *Both chlorophylls absorb most in violet-blue (400-470 nm) and orange-red (630-660 nm); carotenoids at 430-500 nm. Differences between Chl a and b come from their different molecular formulae, structure and colour.*

Action spectrum	***Absorption spectrum***
Effectiveness of wavelengths for photosynthesis	*Absorption of wavelengths by one pigment*

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Action spectrum	Absorption spectrum
Measured by rate of photosynthesis (CO_2 used / O_2 released)	Measured by spectrophotometer
Peaks at blue 430 nm and red 670 nm	Differs for each pigment

SLO 5: Photosystems

- Pigment clusters on the thylakoid membrane = **photosystems**. **Antenna complex** (peripheral): accessory pigments Chl b + carotenoids, gather light. **Reaction centre** (central): only Chl a + proteins; converts energy.

Photosystem I (PS I)	Photosystem II (PS II)
Reaction centre Chl a = P700 (absorbs max 700 nm)	Reaction centre Chl a = P680 (absorbs max 680 nm)
Electrons go to $NADP^+$ (non-cyclic) or return to P700 (cyclic)	Electrons come from photolysis of water; passed to PS I
Used in both cyclic and non-cyclic	Used only in non-cyclic

- Named by order of discovery, not by order in the thylakoid membrane. Excited Chl a electrons are accepted by the first molecule of an electron transport chain and Chl a becomes oxidized.

Overall mechanism of photosynthesis

Light dependent phase (light reaction)	Light independent phase (dark reaction)
In thylakoid membranes (grana); needs light	In stroma; goes on day and night but needs ATP and NADPH made in day
Makes ATP (assimilating power) and NADPH (reducing power)	CO_2 is input; output is glyceraldehyde-3-phosphate (G3P)
Water input, O_2 output	ATP gives energy, NADPH gives electrons and hydrogen

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SLO 6: Non-cyclic and cyclic photophosphorylation

- *ATP production in the light reaction = **photophosphorylation** by **chemiosmosis**.*

(a) Non-cyclic photophosphorylation (Z-scheme)

- *Predominant pathway of higher plants; uses both PS II and PS I and two electron transport chains.*

1. ***Light absorption by PS II:** antenna passes energy to P680; two electrons excited, captured by primary electron acceptor, leaving two electron holes (strong oxidizing agent).*
2. ***Photolysis of water:** H_2O to $2H^+ + 2e^- + 1/2\ O_2$; electrons fill the holes of PS II.*
3. ***PS II to PS I:** electrons go primary acceptor to plastoquinone (PQ), cytochrome b_6-f complex (also a proton pump: H^+ from stroma to thylakoid space), then plastocyanin (PC). H^+ gradient activates ATP synthase: ADP + Pi to ATP (chemiosmosis).*
4. ***Light absorption by PS I:** P700 excited; electrons to primary acceptor; holes filled by electrons from PS II.*
5. ***PS I to NADPH:** electrons to ferredoxin (Fd, iron protein), then NADP reductase: $NADP^+ + 2e^- + 2H^+$ to NADPH + H^+.*

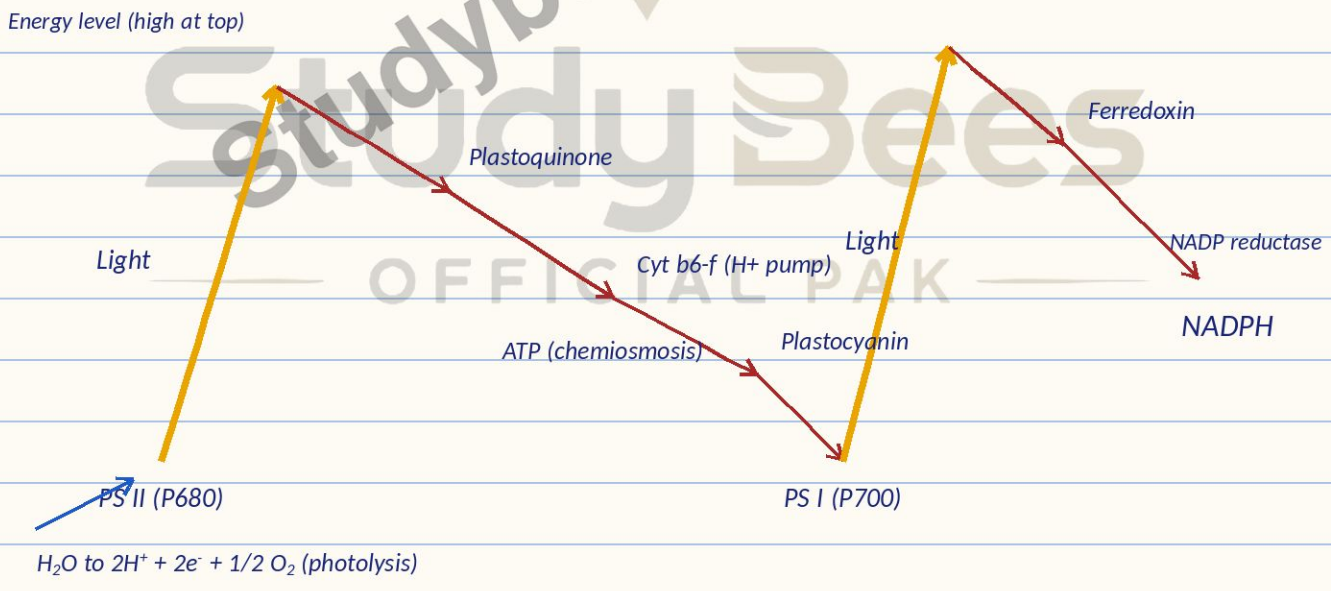


Fig: Z-scheme of non-cyclic photophosphorylation

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(b) Cyclic photophosphorylation

- Occurs when NADPH is high and ATP is short. Only PS I is used. Excited electrons go to primary acceptor, ferredoxin, cytochrome b_6-f complex (ATP made by chemiosmosis) and plastocyanin, then **return to P700. No NADPH, no photolysis of water, no O_2 release.**

Non-cyclic	Cyclic
PS I and PS II	Only PS I
Electrons do not return to the photosystem	Electrons return to P700
ATP + NADPH + O_2 produced	Only ATP
Photolysis of water occurs	No photolysis of water

SLO 7: Calvin cycle (light independent phase)

- Reduction of CO_2 to G3P; worked out by **Melvin Calvin** (Nobel Prize 1961). Also called **C3 pathway** since the first stable compound is 3-carbon 3-PGA. Plants using only Calvin cycle are C3 plants.

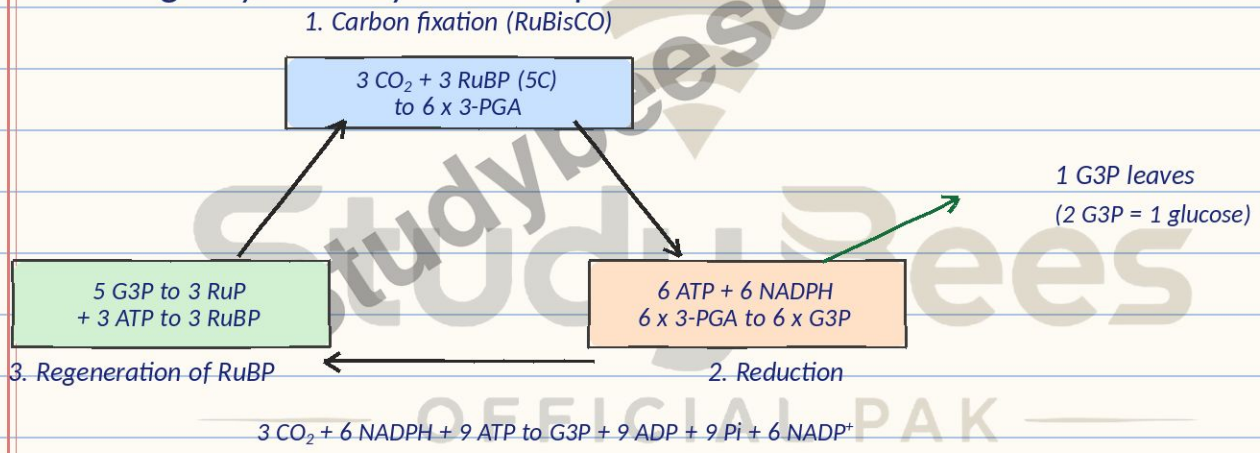


Fig: Calvin cycle (C3 pathway)

1. **Carbon fixation:** CO_2 combines with 5-carbon **RuBP** (CO_2 acceptor) by **RuBisCO** (ribulose bisphosphate carboxylase / oxygenase). Unstable 6-carbon intermediates split into two 3-PGA each.
2. **Reduction:** 6 3-PGA take phosphate from 6 ATP to form 1,3-bisphosphoglycerate, then reduced by NADPH to 6 G3P (phosphate lost). One G3P leaves the cycle; five are used for regeneration.
3. **Regeneration of RuBP:** 5 G3P (15C) rearranged to 3 ribulose phosphate (5C); 3

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ATP convert them to 3 RuBP.

- *For one G3P released: 3 CO_2, 6 NADPH and 9 ATP. Two G3P are needed to make one glucose.* ***3CO_2 + 6NADPH + 9ATP to G3P + 3H_2O + 9ADP + 9Pi + 6$NADP^+$***

4.2 Cellular respiration

Cellular respiration: *A series of complex oxidation-reduction reactions by which living cells obtain energy through breakdown of organic matter (glucose = respiratory fuel).*

- *Aerobic (oxygen present): CO_2 + H_2O, large energy,* ***36 ATP*** *(eukaryotes) or* ***38 ATP*** *(prokaryotes). Anaerobic (no oxygen): lactic acid or ethanol + CO_2, only* ***2 ATP****. Common first step =* ***glycolysis*** *giving 2 pyruvate.*

SLO 8: Anaerobic respiration (fermentation)

- *Incomplete breakdown of glucose without oxygen, in bacteria, yeast, vertebrate muscle and higher plant cells. Pathways: lactic acid fermentation and alcoholic fermentation. NADH gives its electrons to pyruvate / acetaldehyde, so NAD^+ is regenerated for glycolysis to continue.*

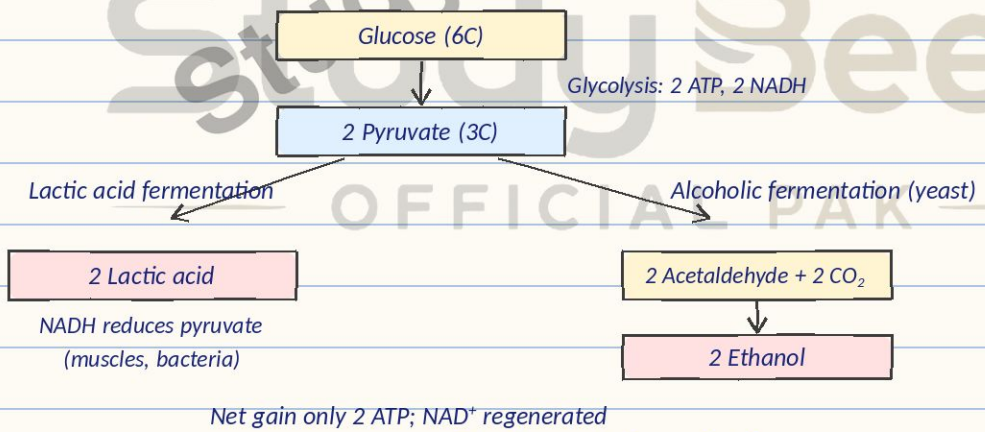


Fig: Anaerobic respiration

- ***Lactic acid fermentation:*** *$C_6H_{12}O_6$ + 2NAD^+ to 2$C_3H_4O_3$ + 2NADH + 2H^+; then 2 pyruvate + 2NADH to 2 lactic acid + 2NAD^+. In anaerobic bacteria and in muscles in strenuous exercise; lactic acid causes muscle fatigue and ache.*
- ***Alcoholic fermentation (yeast):*** *glycolysis, then decarboxylation of pyruvate to*

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acetaldehyde (CO_2 released), then NADH reduces acetaldehyde to ethanol: $2C_3H_4O_3 + 2NADH + 2H^+$ to $2C_2H_5OH + 2CO_2 + 2NAD^+$.

SLO 8: Glycolysis (cytoplasm)

Glycolysis: *Breakdown of glucose (or similar hexose) into 2 pyruvate through enzymatic reactions releasing some ATP and NADH. Occurs in cytoplasm in two phases: preparatory (investment: 2 ATP used) and oxidative (pay-off).*

Step	Reaction and enzyme
1. Phosphorylation	Glucose + ATP to glucose-6-phosphate (hexokinase / glucokinase).
2. Isomerization	G6P to fructose-6-phosphate (phosphohexose isomerase).
3. Phosphorylation	F6P + ATP to fructose-1,6-bisphosphate (phosphofructokinase).
4. Splitting	F-1,6-bisP to G3P + dihydroxyacetone phosphate (aldolase).
5. Isomerization	DHAP to G3P (phosphotriose isomerase). Preparatory phase ends; steps 6-10 occur twice.
6. Dehydrogenation + phosphorylation	G3P + NAD^+ + Pi to 1,3-bisphosphoglycerate + NADH (G3P dehydrogenase).
7. ATP formation	1,3-BPG to 3-phosphoglycerate; ATP (phosphoglycerate kinase), substrate level.
8. Isomerization	3-PG to 2-PG (phosphoglycerate mutase).
9. Dehydration	2-PG to phosphoenolpyruvate (PEP) (enolase).
10. ATP formation	PEP to pyruvate; ATP (pyruvate kinase), substrate level.

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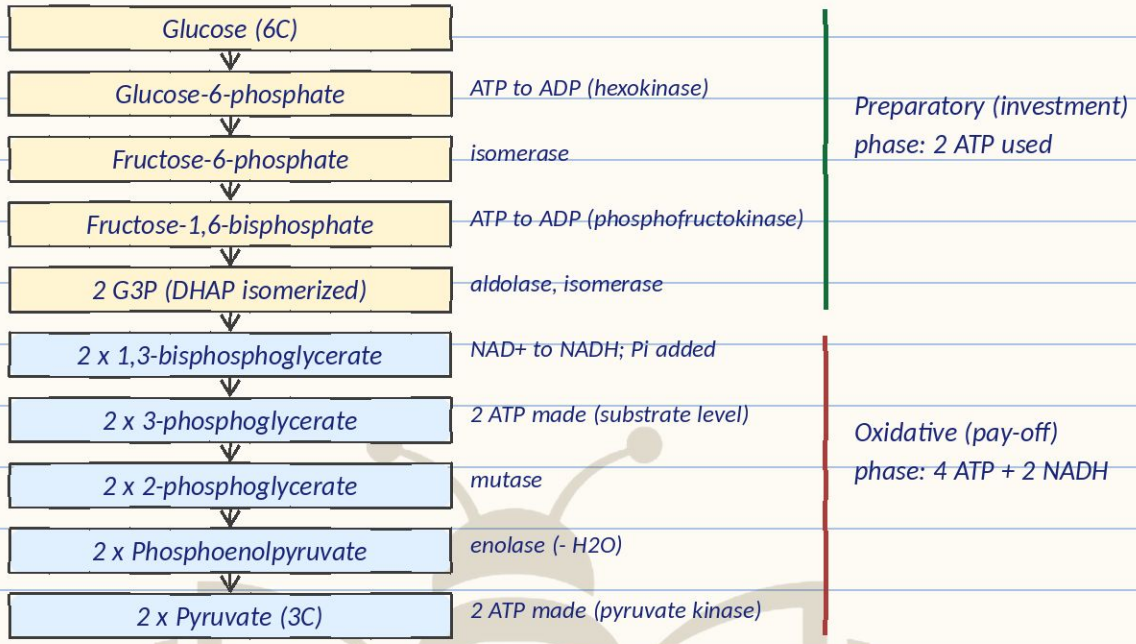


Fig: Glycolysis (cytoplasm); net 2 ATP, 2 NADH, 2 pyruvate

- *Net: 4 ATP made - 2 used = **2 ATP**, 2 NADH, 2 pyruvate.*

SLO 9: Link reaction (oxidation of pyruvate)

- *Pyruvate (charged) enters mitochondrion by active transport through **pyruvate translocase**. The link reaction connects glycolysis (outside) with Krebs cycle (inside). Catalysed by **pyruvate dehydrogenase complex** (3 enzymes).*

1. ***Decarboxylation:** CO_2 removed from pyruvate to give acetaldehyde.*
2. ***Oxidation (dehydrogenation):** NAD^+ removes hydrogen; 2C acetyl group + NADH formed.*
3. ***Addition of coenzyme A:** acetyl + CoA to **acetyl CoA**.*

SLO 10: Krebs cycle (matrix of mitochondrion)

- *Discovered by Sir Hans Krebs. Also **citric acid cycle** or **TCA cycle** (first compound citrate has three carboxylic groups).*

Step	Reaction
1. Synthesis	Acetyl CoA (2C) + oxaloacetate (4C) + H_2O to citrate (6C) (citrate synthetase); CoA freed.
2. Dehydration	Citrate to cis-aconitate (aconitase), loses water.

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Step	Reaction
3. Hydration	cis-Aconitate to isocitrate (aconitase), adds water.
4. Oxidative decarboxylation	Isocitrate to alpha-ketoglutarate (5C) (isocitrate dehydrogenase); NADH + CO_2.
5. Oxidative decarboxylation + CoA	Alpha-ketoglutarate to succinyl CoA (4C) (alpha-KG dehydrogenase); NADH + CO_2.
6. Substrate level ATP	Succinyl CoA to succinate (succinyl thiokinase); GTP then ATP.
7. Dehydrogenation	Succinate to fumarate (succinate dehydrogenase); FAD to $FADH_2$.
8. Hydration	Fumarate to malate (fumarase).
9. Dehydrogenation	Malate to oxaloacetate (malate dehydrogenase); NADH. Cycle repeats.

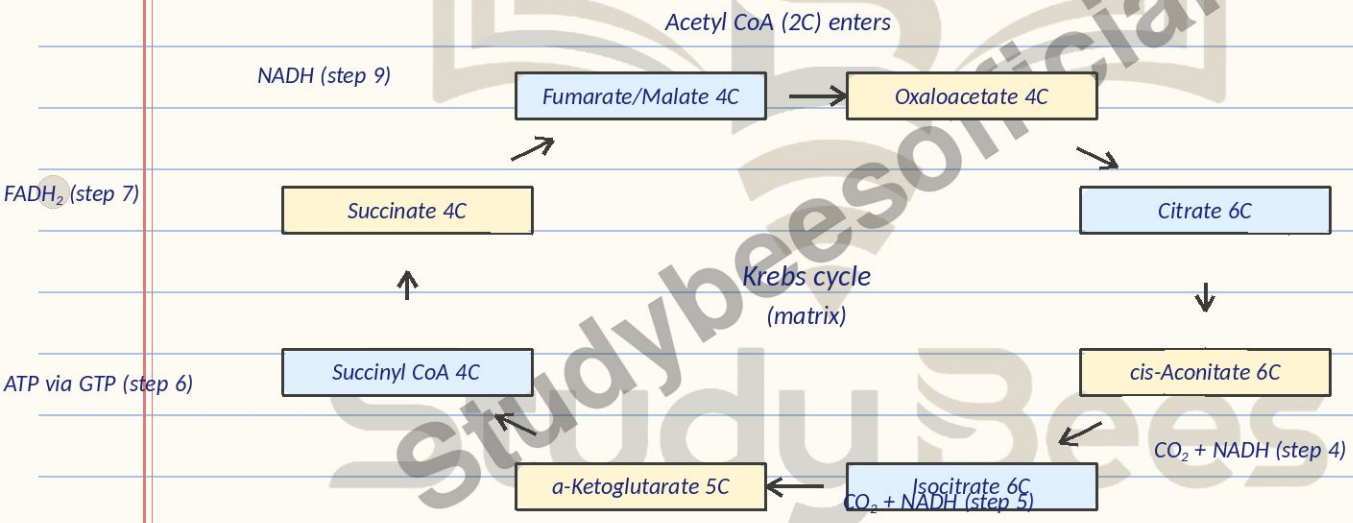


Fig: Krebs cycle (per acetyl CoA: 3 NADH, 1 $FADH_2$, 1 ATP, 2 CO_2)

- **Outputs per glucose (2 turns):** 4 CO_2, 6 NADH, 2 $FADH_2$, 2 ATP.

SLO 11: Electron transport chain (ETC)

- NADH and $FADH_2$ are oxidized and their electrons pass through a series of redox reactions to the final acceptor, **oxygen**. The chain makes no ATP directly; it eases the fall of electrons to oxygen in manageable steps (no energy lost as heat).

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Component	Details
Complex I	NADH-Q reductase (NADH dehydrogenase) with FMN and Fe-S; pumps protons.
Complex II	FADH dehydrogenase; passes electrons from $FADH_2$ to coenzyme Q.
Coenzyme Q (ubiquinone)	Mobile carrier; lipid soluble small molecule, not protein.
Complex III	Cytochrome reductase (Cyt b, Fe-S, Cyt c_1); pumps protons.
Cytochrome c	Mobile electron carrier.
Complex IV	Cytochrome oxidase (Cyt a, a_3, copper); carrier and proton pump; gives electrons to O_2.

- Energy from electron flow pumps H^+ from matrix to intermembrane space at three sites (complexes I, III, IV). Oxygen accepts electrons and picks up H^+ to form **water**. Cytochromes = "cell colour"; reduced cytochromes are pink.

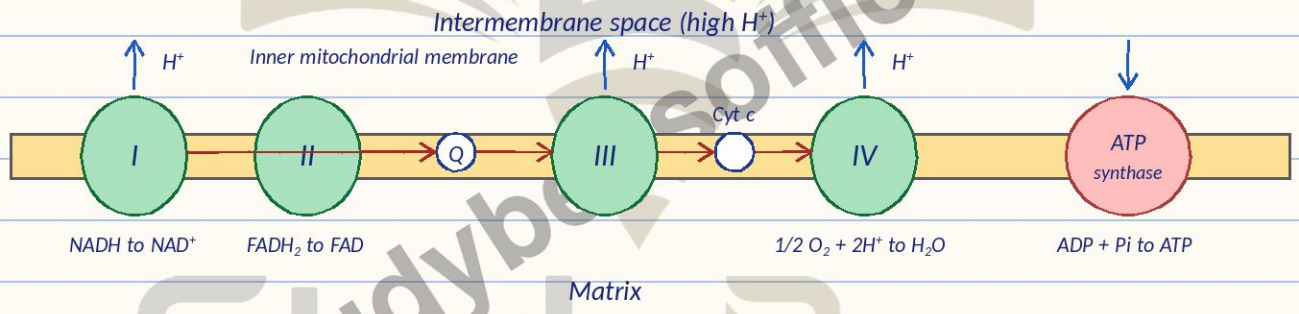


Fig: Respiratory electron transport chain and chemiosmosis

SLO 12: Chemiosmosis and oxidative phosphorylation

Oxidative phosphorylation: Synthesis of ATP using energy liberated by oxidation of reduced coenzymes ($NADH$, $FADH_2$) in the presence of O_2.

- Enzyme **ATP synthetase** in inner mitochondrial membrane has two parts: F_0 (in membrane, proton channel) and F_1 (knob projecting into matrix, catalyses ADP + Pi to ATP).
- Pumping makes high H^+ concentration in the intermembrane space. When protons flow back through F_0, F_1 makes ATP. This coupling = **chemiosmosis**.

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*One NADH gives **3 ATP**; one $FADH_2$ gives **2 ATP**.*

SLO 13: Substrate level phosphorylation

***Substrate level phosphorylation:** Enzymatic transfer of a phosphate from an organic substrate directly to ADP to make ATP. It couples exergonic (energy releasing) reactions with ATP synthesis; the ATP then drives endergonic reactions (e.g. protein synthesis).*

- *Occurs at 3 reactions: glycolysis (1,3-BPG to 3-PG and PEP to pyruvate; 4 ATP, net 2) and Krebs cycle (succinyl CoA to succinate; 2 ATP).*
- ***ATP budget (human cells): 36 ATP = 4 by substrate level + 32 by chemiosmosis.***

Stage	ATP
Glycolysis: net 2 ATP + 2 NADH (cytoplasmic NADH transfers electrons to FAD: 4 ATP)	2 + 4 = 6
Link reaction: 2 NADH	6
Krebs cycle: 2 ATP + 6 NADH (18) + 2 $FADH_2$ (4)	24
Total (eukaryote)	36 (prokaryote 38)

- *Mitochondrial membrane is impermeable to NADH, so eukaryotes yield 2 ATP less than prokaryotes.*

SLO 14: Importance of G3P

1. *Important intermediate of both respiration and photosynthesis.*
2. *In Calvin cycle G3P is converted to glucose phosphate in the chloroplast, then to glucose, fructose, sucrose, maltose and starch.*
3. *In glycolysis it links the preparatory and pay-off phases; ATP and NADH can only be made after G3P forms.*
4. *Leads to pyruvate, the raw material for aerobic and anaerobic respiration.*
5. *Intermediate for inter-conversion of biomolecules, ATP and waste.*
6. *Carbohydrates convert to lipids and lipids to carbohydrates through G3P.*
7. *Carbohydrates convert to amino acids / proteins and back through G3P*

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(metabolic pool).

SLO 15, 16: Respiration of fats and proteins

- ***Fat:*** *hydrolysed to glycerol + three fatty acids. Glycerol converts to* ***G3P*** *(enters glycolysis). Fatty acids convert to* ***acetyl CoA*** *which enters the Krebs cycle (an 18-carbon fatty acid gives 9 acetyl CoA).*
- ***Protein:*** *hydrolysed to amino acids; R-group size decides whether carbon chain is oxidized in glycolysis or Krebs cycle. In the liver deamination removes the amino group, which becomes ammonia and then urea (urea cycle).*

Respiration of glucose	***Respiration of fats***
Glucose breaks into 2 pyruvate by glycolysis	*Fat breaks into glycerol and fatty acids*
Pyruvate to acetyl CoA by link reaction	*Fatty acids directly to acetyl CoA; glycerol to G3P*
Starts with glycolysis in cytoplasm	*Enters at G3P (glycolysis) and acetyl CoA (Krebs)*
Then Krebs cycle and ETC	*Then Krebs cycle and ETC*

4.3 Photorespiration (SLO 17-20)

Photorespiration: *Respiratory activity in green cells in the presence of light that releases carbon dioxide. It needs O_2 and produces CO_2 and H_2O like aerobic respiration, but* ***no ATP is produced****.*

- *When CO_2 in the leaf drops to about 50 ppm, RuBisCO adds O_2 to RuBP instead of CO_2: one phosphoglycerate + one toxic 2C* ***phosphoglycolate*** *(instead of two PGA).*

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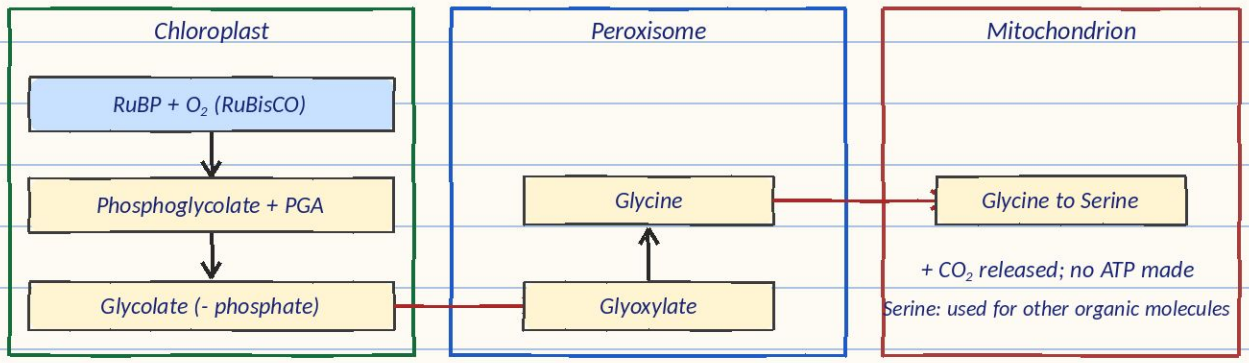


Fig: Photorespiration (chloroplast, peroxisome, mitochondrion)

- *Events: phosphoglycolate loses phosphate to give **glycolate** (chloroplast); transported to **peroxisome**: glyoxylate then **glycine**; glycine goes to **mitochondria** where it becomes **serine**, used to make other organic molecules (CO_2 is released).*

Disadvantages and evolution

- *Costs the plant energy and causes net loss of CO_2 fixation; reduces carbon fixed by about 25% in most plants (up to 5% of photosynthetic potential lost under certain conditions); lowers dry matter and agricultural yield in tropical climates. It is not essential: if inhibited the plant still grows, and some plants are naturally resistant.*
- ***Evolution:** RuBisCO active site evolved to bind both CO_2 and O_2. Originally there was no oxygen in the atmosphere, so only CO_2 binding occurred; photorespiration began when oxygen accumulated.*

SLO 20: Effect of temperature on RuBisCO

- *RuBisCO activity depends on relative O_2 and CO_2 in the leaf. Photorespiration starts when leaf CO_2 is low. In hot conditions stomata close, so CO_2 is used up while O_2 from photosynthesis cannot escape: the O_2 : CO_2 ratio rises and RuBisCO acts as oxygenase (more photorespiration).*

SLO 21: C4 photosynthesis (Hatch-Slack cycle)

- *An adaptation in tropical plants (maize, sugarcane, sorghum) to the problem of photorespiration; extra pathway beside the Calvin cycle.*
- *In cytoplasm of **mesophyll** cells **PEP carboxylase** (not RuBisCO) fixes CO_2 to PEP*

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*(3C) to give **oxaloacetate (4C)**, so these are C4 plants. Oxaloacetate to **malate** (using NADH) is moved to chloroplasts of **bundle sheath** cells; malate to pyruvate with release of CO_2.*

- *CO_2 concentration rises in bundle sheath cells, so RuBisCO works in the Calvin cycle and not in photorespiration. Sugar goes into phloem. Pyruvate returns to mesophyll and regenerates PEP using 1 ATP.*

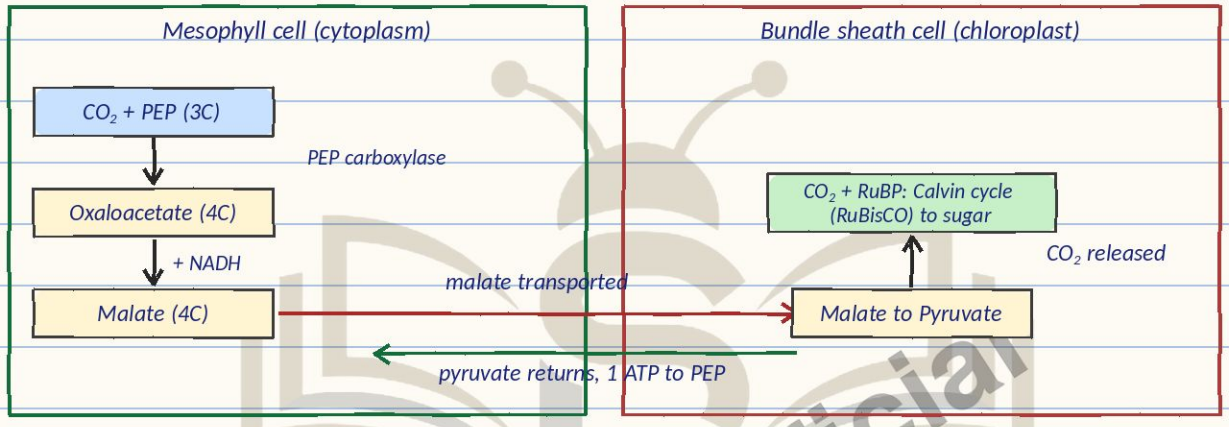


Fig: C4 pathway (Hatch-Slack cycle)

C3 fixation	***C4 fixation***
Enzyme RuBisCO	*Enzyme PEP carboxylase (then RuBisCO in bundle sheath)*
First stable product 3-PGA (3C)	*First product oxaloacetate (4C)*
Only Calvin cycle; mesophyll chloroplasts	*Hatch-Slack + Calvin cycle; mesophyll and bundle sheath*
Photorespiration occurs	*Photorespiration reduced / absent*

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Chapter 4: Short Questions with Answers (SLO based)

Q1. [SLO 1] Define bioenergetics.

Ans: Quantitative study of energy conversions and energy relationships in biological systems.

Q2. [SLO 1] What is photosynthesis chemically and bio-energetically?

Ans: Chemically it is a redox process: CO_2 is reduced to glucose and water is oxidized to oxygen. Bio-energetically it converts energy poor CO_2 and H_2O into energy rich glucose using sunlight.

Q3. [SLO 1] What is the electromagnetic spectrum?

Ans: The full range of electromagnetic radiation in the universe. Visible light (380-750 nm) is a small part, and it drives photosynthesis.

Q4. [SLO 1] Explain the action spectrum of photosynthesis.

Ans: A graph showing effectiveness of different wavelengths of light for photosynthesis (rate of photosynthesis in each colour). Blue (430 nm) and red (670 nm) are most effective.

Q5. [SLO 1] What is the role of carbon dioxide in photosynthesis?

Ans: It is the carbon source used in the light independent reaction (Calvin cycle) to make organic compounds. Rate rises with CO_2 but above 1% it falls due to stomatal closing.

Q6. [SLO 1] What is the role of water in photosynthesis?

Ans: It is the hydrogen and electron donor. Photolysis gives $2H^+$, $2e^-$ and $1/2\ O_2$; electrons refill P680; H^+ and e^- reduce $NADP^+$ to NADPH; oxygen is released.

Q7. [SLO 1] How was it confirmed that plants split water to get hydrogen and release oxygen?

Ans: Van Niel proposed it in 1931. In 1940 water tagged with ^{18}O gave radioactive oxygen, while CO_2 tagged with ^{18}O gave oxygen without the isotope, so oxygen

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comes from water.

Q8. [SLO 2] What are photosynthetic pigments? Name the two major groups in higher plants.

Ans: Substances that absorb light energy for food synthesis. Higher plants have chlorophylls and carotenoids, embedded in thylakoid membranes.

Q9. [SLO 2] What are the types of chlorophyll?

Ans: Chlorophyll a (bluish green, all photosynthetic organisms except bacteria) and chlorophyll b (yellowish green, absent in brown and red algae and bacteria); also Chl c, d, e in algae and bacteriochlorophyll in bacteria.

Q10. [SLO 2] Differentiate between chlorophyll a and chlorophyll b.

Ans: Chl a is bluish green with a methyl group ($-CH_3$) on ring 2, formula $C_{55}H_{72}O_5N_4Mg$. Chl b is yellowish green with an aldehyde group (-CHO), formula $C_{55}H_{70}O_6N_4Mg$.

Q11. [SLO 2] What is the importance of carotene?

Ans: Carotenes are accessory pigments: they absorb light and transfer it to chlorophyll a, protect chlorophyll from excess light, and attract insects and birds for pollination and dispersal.

Q12. [SLO 2] Differentiate between carotene and xanthophyll.

Ans: Carotenes are orange-red isoprenoid pigments in all photosynthetic eukaryotes (beta carotene). Xanthophylls are yellow (lutein, which gives autumn colour).

Q13. [SLO 2] What is chlorosis? Why do leaves turn yellow in autumn?

Ans: Chlorosis is deficiency of chlorophyll. In autumn / winter chlorophyll is not made (needs sunlight, N, Mg^{2+}, warmth) so yellow carotenoids like lutein show.

Q14. [SLO 3] What is the role of photosynthetic pigments?

Ans: They absorb electromagnetic radiation and pass photon energy to the reaction centre; chlorophyll a converts it to chemical energy (ATP and NADPH). Accessory pigments widen the range of light used.

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Q15. [SLO 3] Why are chlorophyll b, xanthophyll and carotene called accessory pigments?

Ans: They absorb light but cannot convert it into chemical energy; they transfer the energy to chlorophyll a.

Q16. [SLO 4] Describe absorption spectrum in photosynthesis.

Ans: A graph showing relative absorption of different wavelengths by a particular pigment, measured with a spectrophotometer.

Q17. [SLO 4] Differentiate between the absorption spectrum of chlorophyll a and b.

Ans: Both absorb mostly violet-blue (400-470 nm) and orange-red (630-660 nm), but peaks differ in size and position because of different molecular formulae, structure and colour.

Q18. [SLO 4] Differentiate between action spectrum and absorption spectrum.

Ans: Action spectrum: effectiveness of wavelengths for photosynthesis. Absorption spectrum: absorption of wavelengths by one pigment.

Q19. [SLO 5] What is a photosystem? Explain.

Ans: A cluster of pigment molecules on the thylakoid membrane. The antenna complex (Chl b, carotenoids) gathers light; the reaction centre (Chl a + proteins) converts it.

Q20. [SLO 5] Differentiate between antenna complex and reaction centre.

Ans: Antenna complex is peripheral, has accessory pigments and absorbs light. Reaction centre is central, has only chlorophyll a and converts light energy into excited electrons.

Q21. [SLO 5] Differentiate between photosystem I and II.

Ans: PS I has P700 and passes electrons to $NADP^+$. PS II has P680, uses electrons from water and passes electrons to PS I.

Q22. [SLO 5] Why are P700 and P680 so named?

Ans: Their reaction centre chlorophyll a absorbs maximum light at 700 nm (PS I) and 680 nm (PS II).

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Q23. [SLO 6] What is photophosphorylation?

Ans: ATP production by chemiosmosis in the light reaction.

Q24. [SLO 6] What is photolysis of water?

Ans: Splitting of water in the presence of light and the oxidized PS II into $2H^+$, $2e^-$ and 1/2 O_2.

Q25. [SLO 6] Why is non-cyclic electron flow called the Z-scheme?

Ans: The path of electrons through the two photosystems has a zigzag shape.

Q26. [SLO 6] Differentiate between cyclic and non-cyclic photophosphorylation.

Ans: Non-cyclic uses PS I and II, makes ATP + NADPH + O_2. Cyclic uses only PS I, electrons return to P700, makes only ATP, no O_2.

Q27. [SLO 6] When does cyclic electron flow occur?

Ans: When NADPH rises and ATP is deficient, until the ATP supply catches up.

Q28. [SLO 6] Differentiate between light dependent and light independent reactions.

Ans: Light reaction: thylakoid, needs light, makes ATP and NADPH, releases O_2. Dark reaction: stroma, uses CO_2, ATP and NADPH to make G3P.

Q29. [SLO 7] What is RuBP and RuBisCO?

Ans: RuBP is the 5-carbon CO_2 acceptor (ribulose bisphosphate). RuBisCO is the enzyme ribulose bisphosphate carboxylase / oxygenase that fixes CO_2.

Q30. [SLO 7] Why is the Calvin cycle also called the C3 pathway?

Ans: Its first stable compound is the 3-carbon 3-phosphoglycerate (3-PGA).

Q31. [SLO 7] Name the three phases of the Calvin cycle.

Ans: Carbon fixation, reduction, and regeneration of RuBP.

Q32. [SLO 7] How many CO_2, NADPH and ATP are needed to release one G3P?

Ans: 3 CO_2, 6 NADPH and 9 ATP; two G3P are needed to make one glucose.

Q33. [SLO 7] Differentiate between the Calvin cycle and the Krebs cycle.

Ans: Calvin: in stroma, anabolic, fixes CO_2 to make G3P using ATP and NADPH. Krebs: in mitochondrial matrix, catabolic, oxidizes acetyl CoA to CO_2 making NADH, $FADH_2$, ATP.

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Q34. [SLO 8] What is glycolysis? Where does it occur?

Ans: Breakdown of glucose into 2 pyruvate with net 2 ATP and 2 NADH; occurs in the cytoplasm.

Q35. [SLO 8] Why is the preparatory phase of glycolysis called the investment phase?

Ans: Two ATP are consumed in it; its end product is two G3P.

Q36. [SLO 8] What are the end products of anaerobic respiration?

Ans: Lactic acid (animals, bacteria) or ethanol and CO_2 (yeast), with only 2 ATP.

Q37. [SLO 8] Differentiate between lactic acid fermentation and alcoholic fermentation.

Ans: Lactic: pyruvate reduced by NADH to lactic acid, in muscles and bacteria. Alcoholic: pyruvate decarboxylated to acetaldehyde (CO_2 lost), reduced by NADH to ethanol, in yeast.

Q38. [SLO 8] What causes muscle fatigue?

Ans: Accumulation of lactic acid in strenuous exercise when oxygen supply is exhausted.

Q39. [SLO 9] What is the link reaction? Where does it occur?

Ans: Oxidation of pyruvate to acetyl CoA (decarboxylation, dehydrogenation, addition of CoA) by pyruvate dehydrogenase complex inside the mitochondrion; it links glycolysis with the Krebs cycle.

Q40. [SLO 10] Why is the Krebs cycle called the citric acid or TCA cycle?

Ans: The first compound formed is citrate (citric acid) which has three carboxylic acid groups (tricarboxylic).

Q41. [SLO 10] Where does the Krebs cycle occur? What does it produce per glucose?

Ans: In the mitochondrial matrix. Per glucose: 4 CO_2, 6 NADH, 2 $FADH_2$ and 2 ATP.

Q42. [SLO 10] Name the first and last compounds of the Krebs cycle.

Ans: First product citrate (6C); the cycle regenerates oxaloacetate (4C).

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Q43. [SLO 11] What is the final acceptor of electrons in ETC? What is formed?

Ans: Molecular oxygen; it picks up H^+ to form water.

Q44. [SLO 11] Name the components of the respiratory electron transport chain.

Ans: Complex I (NADH dehydrogenase), complex II (FADH dehydrogenase), coenzyme Q, complex III (cytochrome reductase), cytochrome c, complex IV (cytochrome oxidase).

Q45. [SLO 11] What are the mobile carriers of the respiratory ETC?

Ans: Coenzyme Q (ubiquinone) and cytochrome c.

Q46. [SLO 11] Why is ETC also called oxidative phosphorylation?

Ans: ATP is made from oxidation of NADH or $FADH_2$ in the presence of oxygen.

Q47. [SLO 12] What is chemiosmosis?

Ans: ATP production using the energy of a proton (H^+) gradient: protons pumped to the intermembrane space flow back through F_0, activating F_1 ATP synthetase.

Q48. [SLO 12] Describe the structure of ATP synthetase.

Ans: F_0 is embedded in the inner membrane and is a proton channel; F_1 is a head-like knob projecting into the matrix that joins ADP and Pi.

Q49. [SLO 12] How many ATP are made from one NADH and one $FADH_2$?

Ans: NADH gives 3 ATP; $FADH_2$ gives 2 ATP.

Q50. [SLO 12] Differentiate between oxidative phosphorylation and photophosphorylation.

Ans: Oxidative: in mitochondria, energy from oxidation of NADH / $FADH_2$, needs O_2. Photo: in chloroplasts, energy from light via electron flow, no O_2 needed.

Q51. [SLO 13] What is substrate level phosphorylation?

Ans: Direct transfer of phosphate from an organic substrate to ADP forming ATP (e.g. PEP to pyruvate; succinyl CoA to succinate).

Q52. [SLO 13] Differentiate between oxidative phosphorylation and substrate level phosphorylation.

Ans: Oxidative uses an electron transport chain, proton gradient and ATP synthetase (32 ATP). Substrate level is a direct enzymatic phosphate transfer

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without ETC (4 ATP).

Q53. [SLO 13] Why do eukaryotes yield 2 ATP less than prokaryotes?

Ans: Mitochondrial membrane is impermeable to cytoplasmic NADH; only its electrons enter and reduce FAD, giving 4 instead of 6 ATP.

Q54. [SLO 14] What is the importance of G3P?

Ans: It is an intermediate of both photosynthesis and respiration, makes glucose and other carbohydrates in Calvin cycle, leads to pyruvate, and inter-converts carbohydrates, lipids and amino acids.

Q55. [SLO 15] How do fats enter respiration?

Ans: Glycerol converts to G3P (glycolysis); fatty acids convert to acetyl CoA (Krebs cycle). An 18C fatty acid gives 9 acetyl CoA.

Q56. [SLO 15] What happens to amino acids in respiration?

Ans: Deamination in the liver removes the amino group (to ammonia then urea); the carbon chain enters glycolysis or the Krebs cycle.

Q57. [SLO 17] Define photorespiration.

Ans: Respiratory activity of green cells in the light that releases CO_2, uses O_2, and produces no ATP.

Q58. [SLO 18] Write the outline of photorespiration.

Ans: RuBisCO adds O_2 to RuBP to give phosphoglycerate + phosphoglycolate; to glycolate; in peroxisome to glyoxylate then glycine; in mitochondria glycine to serine with CO_2 release.

Q59. [SLO 19] What are the disadvantages of photorespiration?

Ans: Costs energy, net loss of CO_2 fixation, reduces carbon fixed by about 25%, lowers dry matter and crop yield in tropical climates.

Q60. [SLO 19] How did photorespiration evolve?

Ans: RuBisCO active site evolved to bind both CO_2 and O_2. It was not a problem when atmosphere had no oxygen; it began when oxygen accumulated.

Q61. [SLO 20] What is the effect of temperature on the activity of RuBisCO?

Ans: At high temperature stomata close, CO_2 is used up and O_2 stays inside, so O_2 : CO_2 rises and RuBisCO acts as oxygenase (photorespiration).

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Q62. [SLO 21] What is the Hatch-Slack cycle? Give examples of C4 plants.

Ans: C4 pathway in which PEP carboxylase fixes CO_2 to PEP giving 4-carbon oxaloacetate, in mesophyll cells. Examples: maize, sugarcane, sorghum.

Q63. [SLO 21] Differentiate between C3 and C4 carbon fixation.

Ans: C3: RuBisCO, first product 3-PGA, photorespiration occurs. C4: PEP carboxylase, first product oxaloacetate, CO_2 concentrated in bundle sheath so photorespiration is reduced.

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Chapter 4: Long Questions with Answers (SLO based)

Q1. [SLO 1] What is photosynthesis? Explain the role of light, carbon dioxide and water in photosynthesis.

Ans:

- Photosynthesis: redox process in which CO_2 is reduced to glucose and water is oxidized to O_2, using light energy. $6CO_2 + 12H_2O$ to $C_6H_{12}O_6 + 6O_2 + 6H_2O$.
- **Light:** sunlight is the ultimate energy source; visible light 380-750 nm; action spectrum shows blue (430 nm) and red (670 nm) most effective; energy is trapped to make high energy organic compounds.
- **CO_2:** carbon source in Calvin cycle (0.03-0.04% of air); rate increases with CO_2, falls above 1% due to stomatal closing.
- **Water:** hydrogen and electron donor; photolysis gives $2H^+$, $2e^-$, $1/2\ O_2$; proved by Van Niel (1931) and ^{18}O experiments (1940).

Q2. [SLO 2] Describe the structure of chlorophyll.

Ans:

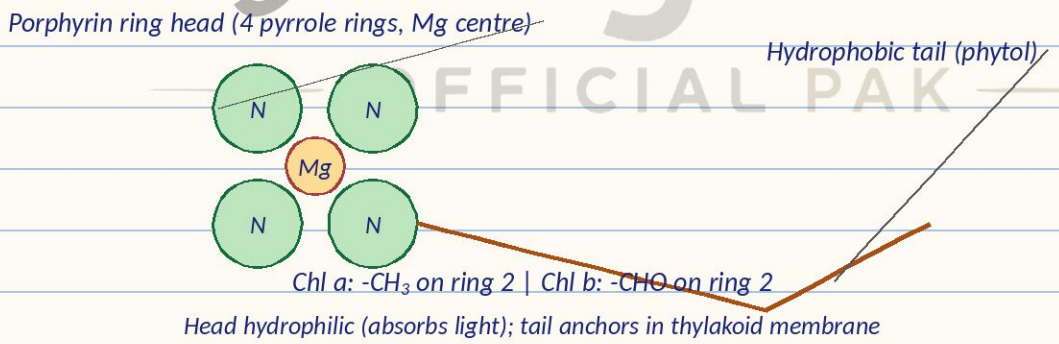


Fig: Structure of chlorophyll

- Chl a = $C_{55}H_{72}O_5N_4Mg$; Chl b = $C_{55}H_{70}O_6N_4Mg$. Molecule = head + two tails.
- Head: porphyrin ring of four pyrrole rings (each 4 C + 1 N); N atoms bond to central Mg; hydrophilic; exposed on thylakoid surface; absorbs light.

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- *Tails (phytol and methyl ester): hydrophobic, embedded in thylakoid membrane, anchor the molecule.*
- *Only difference: Chl a has -CH_3 on ring 2, Chl b has -CHO.*

Q3. [SLO 2] Write a note on the photosynthetic pigment carotene (carotenoids).

Ans:

- *Carotenoids are terpenoid lipids, yellow, orange, red or brown, absorb strongly in blue-violet; seen in leaves before leaf fall, flowers and fruits.*
- *Accessory pigments: absorb light and transfer it to chlorophyll a; protect chlorophyll a from excess light (absorb and disperse); attract insects, birds for pollination and dispersal.*
- *Two types: **carotenes** (orange-red, isoprenoid, in all photosynthetic eukaryotes; beta carotene most important) and **xanthophylls** (yellow; lutein gives autumn leaf colour).*

Q4. [SLO 3, 4] Describe the roles of pigments and differentiate between absorption spectra of chlorophyll a and b.

Ans:

- *Pigments absorb particular wavelengths and reflect the rest (green reflected, so leaves look green). Primary role: absorb radiation and send photon energy to the reaction centre. Chl a converts it to ATP and NADPH; accessory pigments widen the spectrum and protect.*

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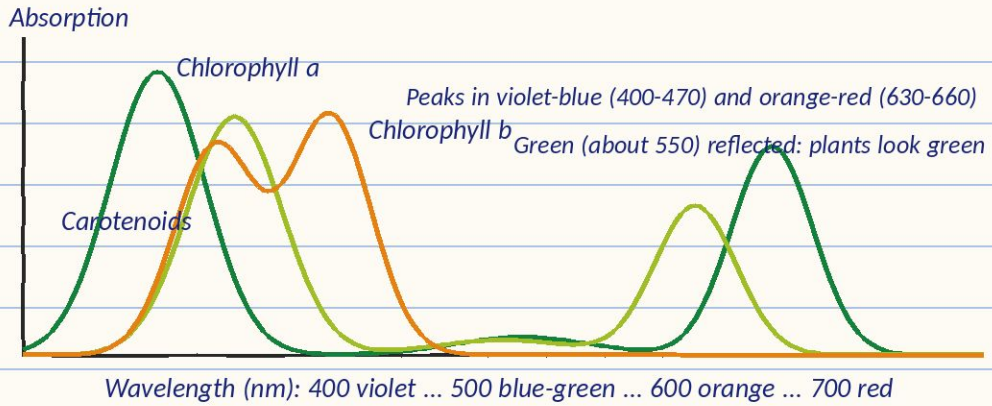


Fig: Absorption spectra of photosynthetic pigments

- *Both chlorophylls absorb most in violet-blue (400-470 nm) and orange-red (630-660 nm); Chl a and b peaks differ due to their molecular formula, structure and colour.*

Q5. [SLO 5] Describe the arrangement of photosynthetic pigments in photosystem I and II.

Ans:

- *Pigments are clustered on thylakoid membranes as photosystems: antenna complex (Chl b, carotenoids) at the periphery and reaction centre (Chl a + proteins) at the centre.*
- *Light energy is passed along antenna pigments to the reaction centre chlorophyll a, whose electrons get excited and escape to a primary electron acceptor.*

PS I	***PS II***
Reaction centre P700	*Reaction centre P680*
Cyclic and non-cyclic	*Non-cyclic only*
Passes electrons to $NADP^+$ / back to P700	*Electrons from water; passes to PS I*

Q6. [SLO 6] Describe the events of non-cyclic photophosphorylation.

Ans:

- *Light absorbed by PS II: P680 excited, electrons to primary acceptor, holes remain.*

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- *Photolysis of water: H_2O to $2H^+ + 2e^- + 1/2\ O_2$; electrons fill PS II holes.*
- *Electron flow PS II to PS I: PQ, cytochrome b_6-f (pumps H^+ into thylakoid space), PC. H^+ gradient drives ATP synthase: ATP made (chemiosmosis).*
- *PS I absorbs light; P700 electrons excited and replaced by those from PS II.*
- *Electron flow PS I to $NADP^+$: ferredoxin, NADP reductase, NADPH formed.*

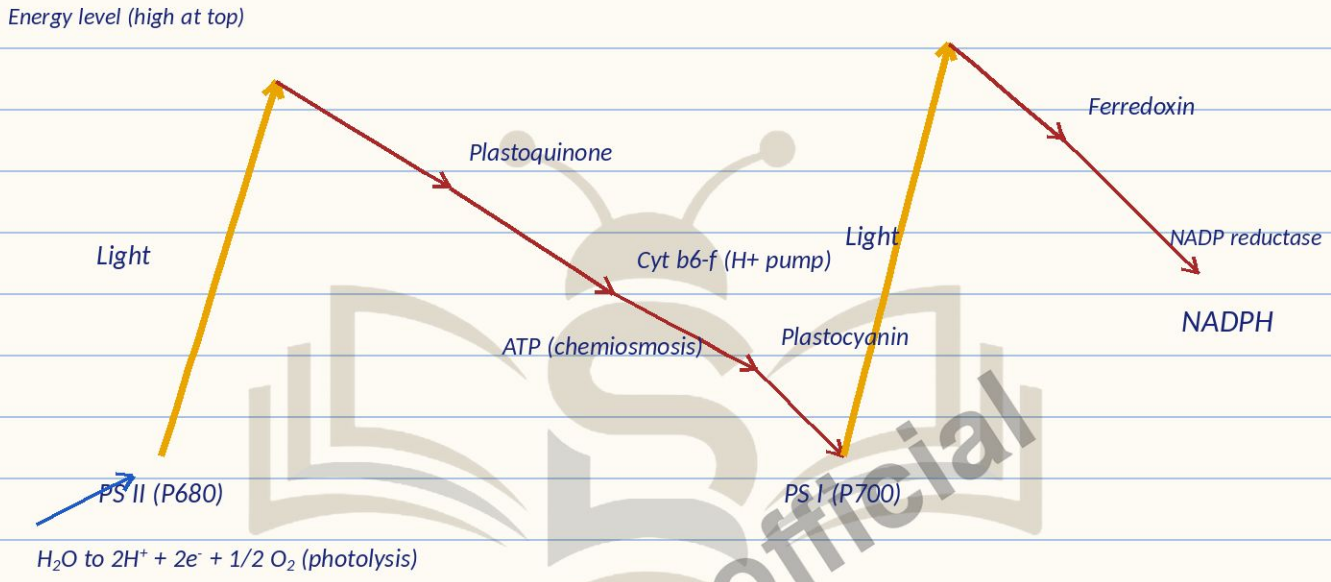


Fig: Z-scheme of non-cyclic photophosphorylation

Q7. [SLO 6] Describe cyclic photophosphorylation and compare it with non-cyclic.

Ans:

- *Only PS I is used. Electrons excited from P700 go to primary acceptor, ferredoxin, cytochrome b_6-f (ATP made by chemiosmosis), plastocyanin and return to P700. No NADPH, no photolysis, no O_2.*

Cyclic	***Non-cyclic***
Only PS I	*PS I and PS II*
Electrons return to P700	*Electrons end in $NADP^+$*
Only ATP	*ATP, NADPH, O_2*
No water splitting	*Photolysis of water*

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Q8. [SLO 7] Explain the Calvin cycle (regeneration of RuBP in outline).

Ans:

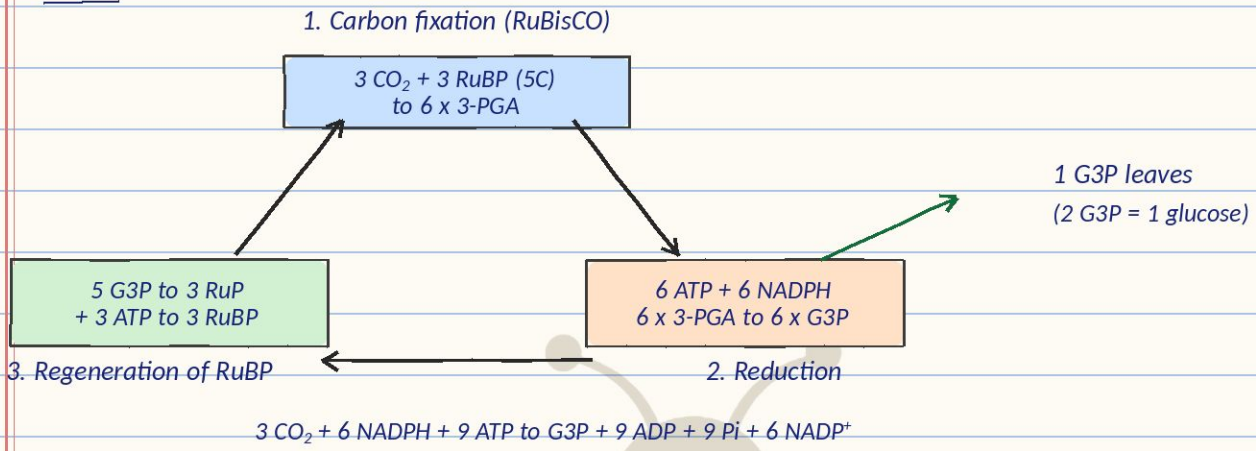


Fig: Calvin cycle (C3 pathway)

- **Carbon fixation:** CO_2 + RuBP by RuBisCO to unstable 6C compounds which split to 3-PGA.
- **Reduction:** 3-PGA + ATP to 1,3-bisphosphoglycerate, reduced by NADPH to G3P; one G3P leaves, five continue.
- **Regeneration:** five G3P to three RuP, then 3 ATP convert to three RuBP.
- Net for one G3P: 3 CO_2, 6 NADPH, 9 ATP.

Q9. [SLO 8] Explain anaerobic respiration in terms of glycolysis and conversion of pyruvate into lactic acid or ethanol.

Ans:

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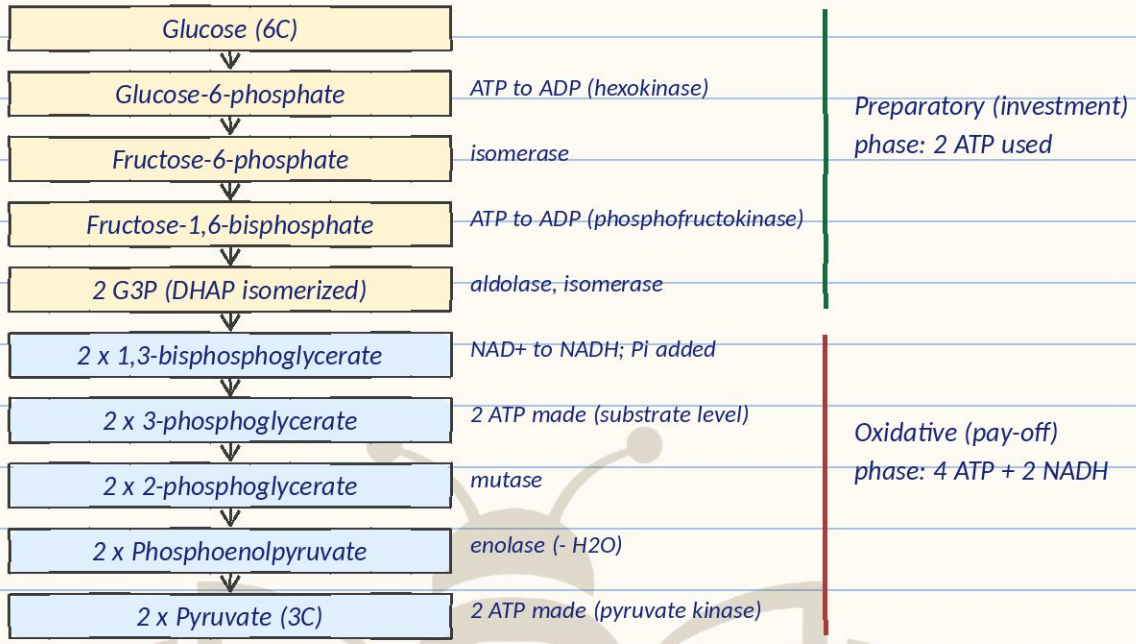


Fig: Glycolysis (cytoplasm); net 2 ATP, 2 NADH, 2 pyruvate

- *Glycolysis (cytoplasm): preparatory phase (2 ATP used, 2 G3P) and oxidative phase (4 ATP, 2 NADH, 2 pyruvate); net 2 ATP.*

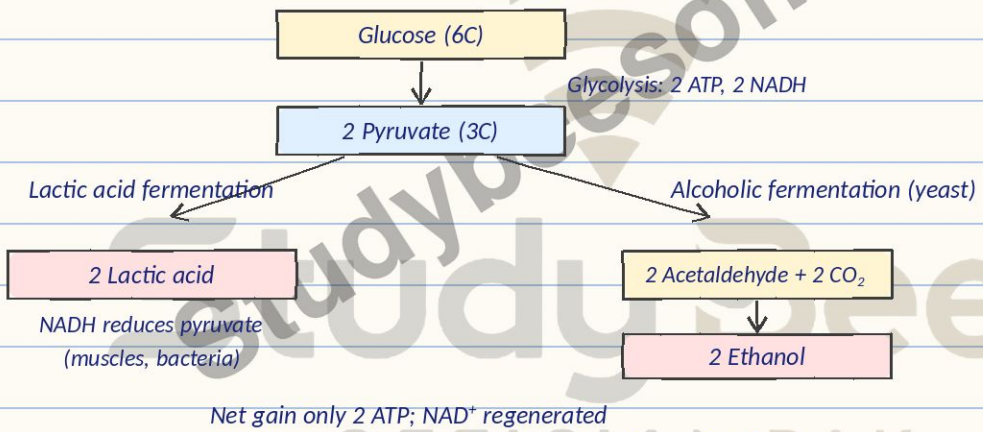


Fig: Anaerobic respiration

- *Lactic acid fermentation: pyruvate + NADH to lactic acid + NAD^+.*
- *Alcoholic fermentation: pyruvate to acetaldehyde + CO_2; acetaldehyde + NADH to ethanol + NAD^+.*
- *NAD^+ is regenerated so glycolysis continues; only 2 ATP per glucose.*

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Q10. [SLO 9] Illustrate the links reaction as conversion of pyruvate to acetyl CoA.

Ans:

- Pyruvate enters mitochondrion by active transport via pyruvate translocase.
- Pyruvate dehydrogenase complex (three enzymes): (1) decarboxylation: CO_2 removed, acetaldehyde; (2) oxidation: NAD^+ removes H, giving 2C acetyl + NADH; (3) acetyl joins coenzyme A to give acetyl CoA.
- Acetyl CoA enters the Krebs cycle.

Q11. [SLO 10] Outline the steps of the Krebs cycle.

Ans:

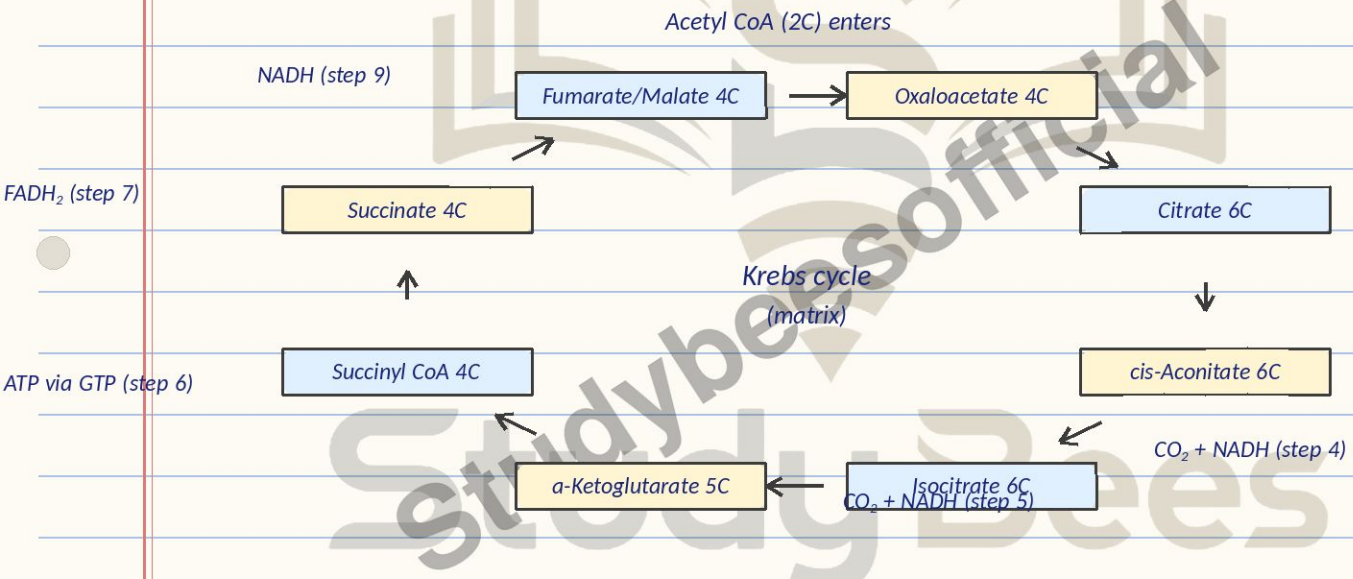


Fig: Krebs cycle (per acetyl CoA: 3 NADH, 1 $FADH_2$, 1 ATP, 2 CO_2)

- Steps: (1) acetyl CoA + oxaloacetate to citrate; (2) dehydration to cis-aconitate; (3) hydration to isocitrate; (4) oxidative decarboxylation to alpha-ketoglutarate (NADH, CO_2); (5) to succinyl CoA (NADH, CO_2); (6) to succinate (GTP / ATP); (7) to fumarate ($FADH_2$); (8) to malate; (9) to oxaloacetate (NADH).
- Per glucose: 4 CO_2, 6 NADH, 2 $FADH_2$, 2 ATP.

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Q12. [SLO 11, 12] Trace the passage of electrons through the ETC and describe chemiosmosis.

Ans:

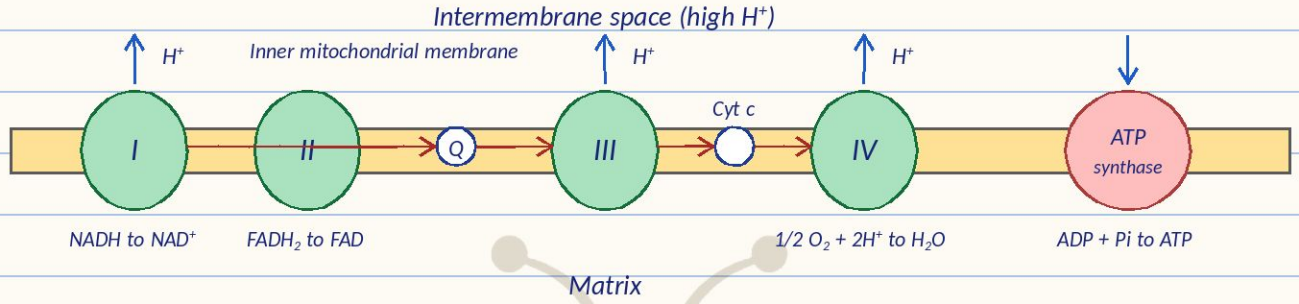


Fig: Respiratory electron transport chain and chemiosmosis

- *NADH gives electrons to complex I (FMN, Fe-S), then to coenzyme Q; $FADH_2$ passes electrons via complex II to Q; Q to complex III, to cytochrome c, to complex IV, finally to O_2 which forms water with H^+.*
- *Energy released at each downhill step pumps H^+ from matrix to intermembrane space at complexes I, III and IV.*
- *Chemiosmosis: H^+ flow back through F_0 activates F_1 of ATP synthetase to make ATP. NADH gives 3 ATP, $FADH_2$ gives 2 ATP.*

Q13. [SLO 13] Explain substrate level phosphorylation and the ATP budget of aerobic respiration.

Ans:

- *Substrate level phosphorylation couples exergonic reactions with ATP synthesis by transferring phosphate from an organic substrate to ADP. It occurs in glycolysis (2 net ATP) and Krebs cycle (2 ATP).*

Stage	ATP
Glycolysis (2 + 4 from NADH electrons)	6
Link reaction	6
Krebs cycle (2 + 18 + 4)	24

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Stage	ATP
Total	36 (4 by substrate level, 32 by chemiosmosis)

Q14. [SLO 14] Justify the importance of G3P in photosynthesis and respiration.

Ans:

- Intermediate of both processes. In Calvin cycle G3P gives glucose phosphate then glucose, fructose, sucrose, maltose and starch.
- In glycolysis it connects preparatory and pay-off phases; ATP and NADH form only after G3P; leads to pyruvate.
- Interconverts carbohydrates, lipids and amino acids / proteins (metabolic pool).

Q15. [SLO 15, 16] Outline the formation of acetyl CoA from fats. Compare respiration of fats and glucose.

Ans:

- Fat to glycerol + 3 fatty acids. Glycerol to G3P (enters glycolysis). Fatty acids to acetyl CoA (enters Krebs cycle); an 18C fatty acid gives 9 acetyl CoA.

Glucose	Fat
Glycolysis to pyruvate to acetyl CoA (link reaction)	Fatty acids directly to acetyl CoA; glycerol to G3P
Acetyl CoA to Krebs and ETC	Acetyl CoA to Krebs and ETC

Q16. [SLO 17-19] Define photorespiration. Outline its events. Explain its disadvantages and evolution.

Ans:

- Photorespiration: respiratory activity in green cells in light, releasing CO_2; needs O_2; no ATP formed.

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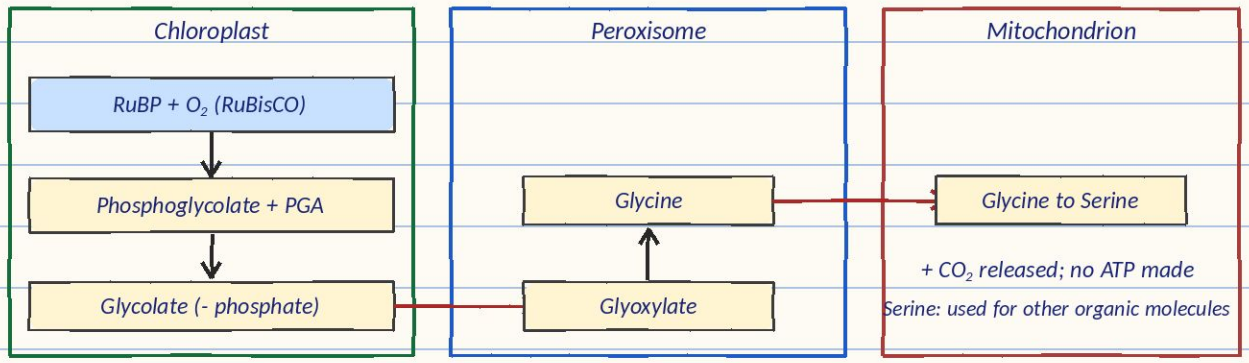


Fig: Photorespiration (chloroplast, peroxisome, mitochondrion)

- *Events: at CO_2 about 50 ppm RuBisCO adds O_2 to RuBP giving PGA + phosphoglycolate; to glycolate; peroxisome: glyoxylate then glycine; mitochondria: glycine to serine.*
- *Disadvantages: costs energy, loses fixed CO_2, reduces carbon fixation by about 25%, reduces yield in tropics; not essential.*
- *Evolution: RuBisCO binds both CO_2 and O_2; it began when oxygen accumulated in the atmosphere.*

Q17. [SLO 20, 21] Explain the effect of temperature on RuBisCO and the C4 photosynthesis as an adaptation.

Ans:

- *High temperature closes stomata; CO_2 falls and O_2 rises in the leaf; RuBisCO acts as oxygenase and photorespiration increases.*

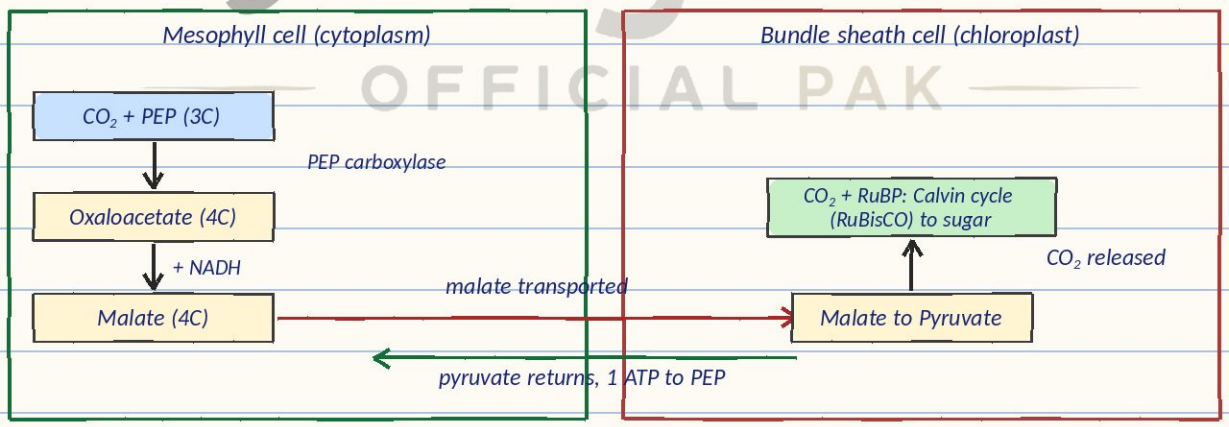


Fig: C4 pathway (Hatch-Slack cycle)

- *C4 plants (maize, sugarcane, sorghum): PEP carboxylase in mesophyll fixes CO_2 to PEP, giving oxaloacetate; malate moves to bundle sheath chloroplasts and*

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releases CO_2; high CO_2 makes RuBisCO work in the Calvin cycle; pyruvate returns and regenerates PEP using 1 ATP.

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